

Q1 - 24 June - Shift 2

If $y = \tan^{-1}(\sec x^3 - \tan x^3)$, $\frac{\pi}{2} < x^3 < \frac{3\pi}{2}$, then

Space for your notes:

(A) $xy'' + 2y' = 0$

(B) $x^2y'' - 6y + \frac{3\pi}{2} = 0$

(C) $x^2y'' - 6y + 3\pi = 0$

(D) $xy'' - 4y' = 0$

Q2 - 25 June - Shift 1

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^3 + x - 5$.
If $g(x)$ is a function such that $f(g(x)) = x$,
 $\forall x \in \mathbb{R}$, then $g'(63)$ is equal to _____.

Space for your notes:

(A) $\frac{1}{49}$

(B) $\frac{3}{49}$

(C) $\frac{43}{49}$

(D) $\frac{91}{49}$

Q3 - 26 June - Shift 2

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(x+y) = 2^x f(y) + 4^y f(x), \forall x, y \in \mathbb{R}$. If $f(2) = 3$, then $14 \cdot \frac{f'(4)}{f'(2)}$ is equal to _____.

Space for your notes:

Q4 - 27 June - Shift 1

If $\cos^{-1}\left(\frac{y}{2}\right) = \log_e\left(\frac{x}{5}\right)^5$, $|y| < 2$, then :

Space for your notes:

(A) $x^2 y'' + xy' - 25y = 0$

(B) $x^2 y'' - xy' - 25y = 0$

(C) $x^2 y'' - xy' + 25y = 0$

(D) $x^2 y'' + xy' + 25y = 0$

Q5 - 27 June - Shift 2

Let $f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$, $a \in \mathbb{R}$. Then the sum of

Space for your notes:

which the squares of all the values of a for

$2f'(10) - f'(5) + 100 = 0$ is :

(A) 117

(B) 106

(C) 125

(D) 136

Q6 - 27 June - Shift 2

If $y(x) = (x^{x^x})$, $x > 0$ then $\frac{d^2 x}{dy^2} + 20$ at $x = 1$ is equal to:

Space for your notes:

Answer Key

Q1 (B)

Q2 (A)

Q3 (248)

Q4 (D)

Q5 (C)

Q6 (16)

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Q1 (B)

$$y = \tan^{-1}(\sec x^3 - \tan x^3)$$

$$= \tan^{-1}\left(\frac{1 - \sin x^3}{\cos x^3}\right)$$

$$= \tan^{-1}\left(\frac{1 - \cos\left(\frac{\pi}{2} - x^3\right)}{\sin\left(\frac{\pi}{2} - x^3\right)}\right)$$

$$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} - \frac{x^3}{2}\right)\right)$$

$$\text{Since } \frac{\pi}{4} - \frac{x^3}{2} \in \left(-\frac{\pi}{2}, 0\right)$$

$$y = \left(\frac{\pi}{4} - \frac{x^3}{2}\right)$$

$$y' = \frac{-3x^2}{2}, y'' = -3x$$

$$4y = \pi - 2x^3$$

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$$12y = 3\pi + 2x^2y''$$

$$x^2y'' - 6y + \frac{3\pi}{2} = 0$$

Q2 (A)

$$f(x) = x^3 + x - 5$$

$$\Rightarrow f'(x) = 3x^2 + 1 \Rightarrow \text{increasing function}$$

$\Rightarrow$ invertible

$\Rightarrow g(x)$ is inverse of $f(x)$

$$\Rightarrow g(f(x)) = x$$

$$\Rightarrow g'(f(x))f'(x) = 1$$

$$f(x) = 63$$

$$\Rightarrow x^3 + x - 5 = 63$$

$$\Rightarrow x = 4$$

put $x = 4$

$$g'(f(4))f'(4) = 1$$

$$g'(63) \times 49 = 1 \quad \{f(4) = 49\}$$

$$g'(63) = \frac{1}{49}$$

Q3 (248)

Put $y = 2$

$$f(x + y) = 2^x \cdot f(y) + 4^y \cdot f(x).$$

$$f(x + 2) = 2^x \cdot 3 + 16f(x)$$

$$f'(x + 2) = 16f'(x) + 3 \cdot 2^x \ln 2$$

$$f'(4) = 16f'(2) + 12 \ln 2 \quad \dots(i)$$

$$f(y + 2) = 4f(y) + 3 \cdot 4^y$$

$$f'(4) = 4f'(y) + 3 \cdot 4^y \ln 4$$

$$f'(4) = 4f'(2) + 96 \ln 2 \quad \dots(ii)$$

solving eq. (i) and (ii), we get

$$f'(2) = 7 \ln 2$$

from equation (i), we get

$$f'(4) = 124 \ln 2$$

$$\text{Now, } \Rightarrow 14 \cdot \frac{f'(4)}{f'(2)}$$

$$14 \times \frac{124 \ln 2}{7 \ln 2}$$

$$= 248.$$

Q4 (D)

$$\cos^{-1}\left(\frac{y}{2}\right) = \log_e\left(\frac{x}{5}\right)^5$$

$$\cos^{-1}\left(\frac{y}{2}\right) = 5 \log_e\left(\frac{x}{5}\right)$$

$$\frac{-1}{\sqrt{1-\frac{y^2}{4}}} \cdot \frac{y'}{2} = 5 \cdot \frac{1}{x} \times \frac{1}{5}$$

$$\Rightarrow \frac{-y'}{\sqrt{4-y^2}} = \frac{5}{x}$$

$$-xy' = 5\sqrt{4-y^2}$$

$$-xy'' - y' = 5 \cdot \frac{1}{2\sqrt{4-y^2}} (-2y \cdot y')$$

$$\Rightarrow xy'' + y' = \frac{5y' \cdot y}{\sqrt{4-y^2}}$$

$$xy'' + y' = 5 \cdot \left(\frac{-5}{x}\right) y$$

$$x^2 y'' + xy' = -25y$$

Q5 (C)

$$f(x) = \begin{vmatrix} a & -1 & 0 \\ ax & a & -1 \\ ax^2 & ax & a \end{vmatrix}$$

$$f(x) = a \begin{vmatrix} 1 & -1 & 0 \\ x & a & -1 \\ x^2 & ax & a \end{vmatrix}$$

$$= a [1 (a^2 + ax) + 1 (ax + x^2)]$$

$$\Rightarrow f(x) = a (x + a)^2$$

$$\text{so, } f'(x) = 2a (x + a)$$

$$\text{as, } 2f'(10) - f'(5) + 100 = 0$$

$$\Rightarrow 2 \times 2a (10 + a) - 2a (5 + a) + 100 = 0$$

$$\Rightarrow 40a + 4a^2 - 10a - 2a^2 + 100 = 0$$

$$2a^2 + 30a + 100 = 0$$

$$\Rightarrow a^2 + 15a + 50 = 0$$

$$(a + 10)(a + 5) = 0$$

$$a = -10 \text{ or } a = -5$$

$$\text{Required} = (-10)^2 + (-5)^2 = 125$$

Q6 (16)

Hints and Solutions

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$$y = (x) = (x^x)^x$$

$$\ln y(x) = x^2 \cdot \ln x$$

$$\frac{1}{y(x)} \cdot y'(x) = \frac{x^2}{x} + 2x \cdot \ln x$$

$$y'(x) = y(x) [x + 2x \ln x]$$

$$y(1) = 1; y'(1) = 1$$

$$y''(x) = y'(x) [x + 2x \cdot \ln(x)] + y(x) [1 + 2(1 + \ln x)]$$

$$y''(1) = 1 [1 + 0] + 1 (1 + 2) = 4$$

$$\frac{d^2 y}{dx^2} = - \left(\frac{dy}{dx} \right)^3 \cdot \frac{d^2 x}{dy^2}$$

$$\Rightarrow 4 = - \frac{d^2 x}{dy^2}$$

$$\frac{d^2 x}{dy^2} = -4$$

$$\text{Ans. } -4 + 20 = 16$$

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